

PROGRAMMING METHODS - LABORATORY 2

Task 1

Two motor models are given:

A. a squirrel-cage asynchronous electric motor described by:

$$K1(s) = k_{p1} * (1 + T_z*s) / ((1 + T1*s) * (1 + T2*s))$$

B. a brushless electric motor described by:

$$K2(s) = k_{p2} / (L*J*s^2 + R*J*s + 2*k_{p2}^2)$$

Identification grids

81 identification grids were generated for the squirrel-cage asynchronous electric motor and saved in the text file siatkaSEAK.txt. A fragment of one grid has the following structure:

	Model	kp	Tz	T1	T2	T	...
Y	5	200	0.5	0.3	0.25	0.606003	0, 0.009858, 0.0192428, ...
X	0	0	0	0	0	0	0, 0.0298957, 0.0589964, ...

The Y row contains 106 columns of data: the first 6 columns are parameters, and the next 100 columns are y-coordinates of vector Y. The X row also contains 106 columns: the first 6 columns are parameters, and the next 100 columns are x-coordinates of vector X.

One identification grid of an unknown model was also generated and saved in nmSEAK.txt. The layout of this file is identical to siatkaSEAK.txt.

81 identification grids were also generated for the brushless electric motor and saved in siatkaSEB.txt. A fragment of one grid has the following structure:

	Model	kp	L	R	J	T	...
Y	3	5	0.0005	0.05	0.001	0.00061	0, 0.0006973, 0.0027851, ...
X	0	0	0	0	0	0	0, 0.0011623, 0.0046446, ...

One identification grid of an unknown model was also generated and saved in nmSEB.txt. The layout of this file is identical to siatkaSEB.txt.

Program to write

Write a program that determines the best match between the unknown identification grid model and the group of 81 identification grids, using the minimum MSE error according to the formula:

$$mse = \sqrt{\sum_{i=0}^{\{n\}} (Y_i - Y_i')^2 + \sum_{i=0}^{\{n\}} (X_i - X_i')^2}$$

Perform this for: A) the squirrel-cage asynchronous electric motor; B) the brushless electric motor.

Divide the program into .cpp and .h files containing thematically organized functions together with function descriptions according to the rules presented during class. Do not forget comments in the code.

Example program structure

The text files with input data are loaded into the main() function, where the whole program is executed. This function is located in Main.cpp. The following header files are included in Main.cpp:

- LibFiles.h - declarations of functions needed for text-file operations.
- LibTables.h - declarations of functions needed for operations on dynamic two-dimensional arrays that store the input data read from text files.

The above headers are also included in LibFiles.cpp and LibTables.cpp, which contain the definitions of the functions declared in the corresponding header files. After the program finishes, the results are saved to a text file.

Decide where to store the declaration and definition of the function/algorithm that determines the best match with minimum MSE. It may be a separate library or one of the existing libraries.

Input

Data are located in text files. Their names are provided by the user while the program is running. The file layout is described above.

Output

A text file with a name provided by the user containing:

- MSE error values for every grid, separated by spaces;
- the minimum MSE error for the best match;
- the Y and X vector with all data of the best match.

Draw a scatter plot showing the grid of the unknown model and the grid of the best-matching model.

Examples of result files

Squirrel-cage asynchronous electric motor: the source example includes 81 MSE values, the minimum MSE equal to 0.131767, and the full Y/X vector of the best fit.

Brushless electric motor: the source example includes 81 MSE values, the minimum MSE equal to 1.18146, and the full Y/X vector of the best fit.

Example plots

The required plots should compare the identification-grid model with the unknown motor model. The first plot is for the squirrel-cage asynchronous motor, and the second plot is for the brushless motor.

Submission rules

The .cpp file(s) should be named:

Surname_Name_Program_01.cpp

The source files, all other created files, and test text files should be placed in the folder Surname_Name_Laboratory_2.

Include all created files, including text files with sample data and plots.

The folder should be packed as a .rar or .zip archive and submitted to the course folder for Programming Methods - laboratory 2 on delta.pk.edu.pl.